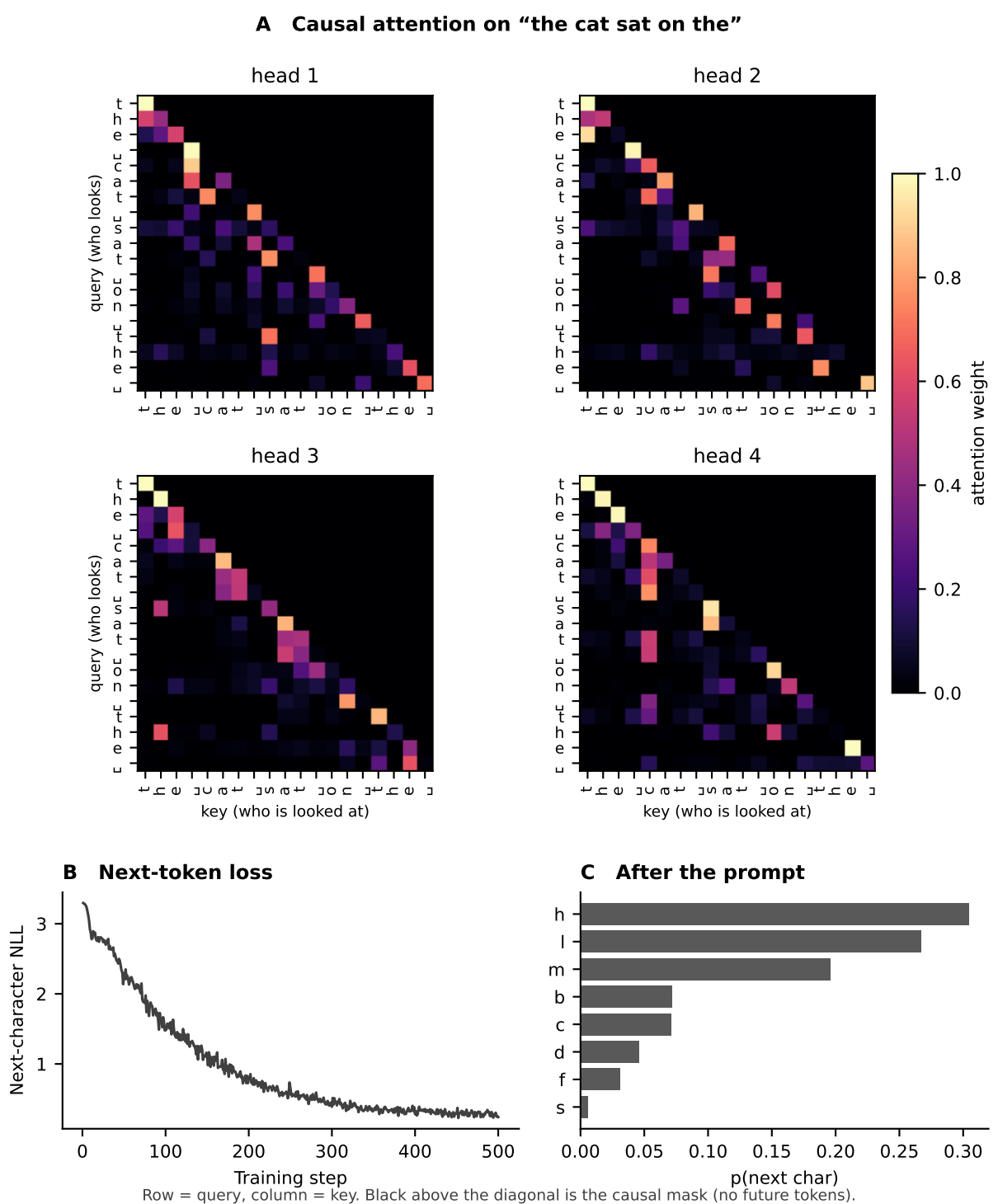
A Causal attention on “the cat sat on the”
head 1
head 2
head 3
head 4
query (who looks)
key (who is looked at)
attention weight
1.0
0.8
0.6
0.4
0.2
0.0
B Next-token loss
Next-character NLL
Training step
C After the prompt
p(next char)
Row = query, column = key. Black above the diagonal is the causal mask (no future tokens).